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Experimental investigation and thermodynamic re-assessment of the Mn–Nd binary system

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Abstract Fourteen Mn–Nd alloys were investigated experimentally by means of thermal analysis. The temperatures of the invariant reactions and liquidus in the Mn–Nd binary system were determined. Based on the experimental results obtained in the present work as well as the experimental data reported and the previous assessments in the literature, the Mn–Nd binary system was re-assessed thermodynamically using the CALPHAD method. The solution phases, including liquid, α-Mn, β-Mn, γ-Mn, δ-Mn, α-Nd and δ-Nd, are modeled by the substitutional solution model, and their excess Gibbs energies are expressed with the Redlich–Kister polynomial. The intermetallic compounds, Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$, are treated as stoichiometric compounds. A set of self-consistent thermodynamic parameters obtained finally to describe the Gibbs energies of various phases in the Mn–Nd binary system can be used to reproduce well the reported phase equilibria and thermodynamic data.

Keywords Mn–Nd · Thermodynamics · Thermal analysis · CALPHAD

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Introduction

Mn-containing ternary compounds with rare-earth metals (RE) and other elements (e.g., Ge, Si, Sn and In) have been investigated extensively in order to understand the relations between their crystal structures and magnetic properties [1–6]. The stable intermetallic compounds in the RE–Mn–Ge (Si, Sn, In) systems show complex magnetic properties, mainly resulting from complex exchange interactions between 4*f*-moments of the rare-earth metals and the itinerant 3*d* moments of Mn [7–9]. In particular, the ternary compounds (e.g., RMn_2X_2 with $ThCr_2Si_2$-typed structure and REMnX compounds with CeFeSi or TiNiSi-typed structure) in the Nd–Mn–Ge/Si ternary systems were discovered to show interesting magnetic properties including magnetocaloric effect, gaint magnetoresistance effect and magnetic-field-induced strain [10–14].

To understand better the intermetallic compounds with desirable physical properties in the Nd–Mn–Ge/Si ternary systems, the knowledge of phase equilibria and thermodynamic properties of the related Nd–Mn–Ge/Si systems is indispensable. As a key subsystem in the Nd–Mn–Ge/Si ternary systems, phase equilibria and thermodynamic properties of the Mn–Nd binary system are required firstly. Based on the reported experimental information, thermodynamic calculation of the Mn–Nd binary system was carried out by Kim and Jung [15] using the quasi-chemical model to describe the liquid phase. Consequently, on the basis of the experimental investigations of phase equilibria, Mostafa et al. [16] re-optimized the Mn–Nd binary system using the same models in Ref. [15] because a new intermetallic compound, $Mn_{17}Nd_2$, was confirmed to be stable at room temperature, except for two stable Mn_2Nd and $Mn_{23}Nd_6$. Although the calculated results in Refs. [15, 16] are in good agreement with the experimental data,

the stability of the intermetallic compounds, Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$, would be determined in further experimental verification. In addition, considering the compatibility of thermodynamic database of the Mn–RE multi-component systems published in the literature, thermodynamic re-assessment of the Mn–Nd binary system is still necessary using the Thermo-calc® software.

Therefore, in the present work, the Mn–Nd binary system was firstly investigated experimentally by means of differential thermal analysis (DTA) and differential scanning calorimeter (DSC), and then thermodynamic re-assessment of this binary system was performed using the CALPHAD [17] method based on the critical review of the reported experimental information and previous assessments.

Literature information

Phase diagram data

The phase diagram of the Mn–Nd binary system was firstly studied experimentally by Kirchmayr and Lugscheider [18] using X-ray diffraction (XRD) and DTA. The temperatures of the liquidus and invariant reactions were determined, while the solubilities of Nd and Mn in the terminal solid solution phases (α-Mn, β-Mn, γ-Mn, δ-Mn, α-Nd and β-Nd) were neglected. Three intermetallic compounds, $Mn_{12}Nd$, $Mn_{23}Nd_6$ and Mn_2Nd, were confirmed. Makhalenko and Kuźma [19] investigated the partial Mn–Nd phase diagram by means of XRD in the composition range from 55 to 95 at.% Mn (773–1073 K). Their experimental results show two intermetallic compounds, $Mn_{23}Nd_6$ and Mn_2Nd, are stable in the limited temperature ranges. Based on the experimental results [18, 19], the liquidus boundaries were redrawn by Okamoto [20]. Subsequently, Saccone et al. [21] investigated the phase diagram of the Mn–Nd binary system using DTA, XRD, scanning electron microscopy (SEM) and electron probe microanalysis (EPMA). The temperatures of liquidus determined in their experiments are lower than those measured by Kirchmayr and Lugscheider [18]. In agreement with the experimental results of Makhalenko and Kuźma [19], two intermetallic compounds, Mn_2Nd and $Mn_{23}Nd_6$, were confirmed and formed pertectically at 1093 and 1203 K and decomposed eutectoidally at about 853 and 753 K, respectively. Moreover, the two peritectic reactions (L + β-Mn ↔ $Mn_{23}Nd_6$ at 1203 K, L + $Mn_{23}Nd_6$ ↔ Mn_2Nd at 1093 K), the catatectic reaction (α-Nd ↔ L + β-Nd at 1136 K), the eutectic reaction (L ↔ Mn_2Nd + α-Nd at 968 K) and two eutectoid reactions ($Mn_{23}Nd_6$ ↔α-Nd + α-Nd at about 753 K and Mn_2Nd ↔ $Mn_{23}Nd_6$ + α-Nd at about 853 K) were recommended by Saccone et al. [21].

On the basis of the experimental information [18, 19, 21], Kim and Jung [15] assumed the solubility of Mn in β-Nd is 4 at.% Mn at 1009 K in the calculation. Recently, Mostafa et al. [16] re-investigated the phase diagram of the Mn–Nd binary system by means of DSC, XRD, SEM and EPMA techniques. The two compounds Mn_2Nd and $Mn_{23}Nd_6$ are confirmed, which are formed at 1123 and 1213 K, respectively. A new intermetallic compound $Mn_{17}Nd_2$ was observed, which was formed through the peritectic reaction (L + β-Mn ↔ $Mn_{17}Nd_2$ at 1298 K). The terminal solid solubility of Mn in α-Nd is 2.3 at.% Mn at 958 K, and the solubility of Mn in β-Nd is 5.0 at.% Mn at 1001 K.

Recently, the Mn–Nd binary system was reviewed by Okamoto [23, 24] based on the latest experimental results determined by Saccone et al. [21] and Mostafa et al. [16] as well as the previous review of Okamoto [20, 22].

Thermodynamic data

Ivanov et al. [25] employed the isoperibolic calorimetry method to measure the enthalpy of mixing of the liquid Mn–Nd alloys and the partial enthalpies of mixing of Nd and Mn at 1600 K. The enthalpies of mixing of the Mn–Nd liquid alloys vary from negative values in the Mn-rich part to positive values in the Nd-rich part. The maximum and minimum values of the enthalpies of mixing of the Mn–Nd liquid alloys are about 1.99 and −0.4 kJ mol^{-1}, respectively.

Up to now, the enthalpies of formation of the intermetallic compounds (Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$) in the Mn–Nd binary system were not determined experimentally in the published literature. In the thermodynamic optimization of the Mn–Nd binary system, the enthalpies of formation of Mn_2Nd and $Mn_{23}Nd_6$, were calculated by Kim and Jung [15] to be 5400 and 3910 J mol^{-1}, respectively. Subsequently, Mostafa et al. [16] used the first-principle calculations to obtain the enthalpies of formation of Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$ to be 3650, 2150 and 1250 J mol^{-1}, respectively. Meanwhile, the enthalpies of formation of Mn_2Nd (4200 J mol^{-1}), $Mn_{23}Nd_6$ (2002 J mol^{-1}) $Mn_{17}Nd_2$ (87 J mol^{-1}) were also assessed by Mostafa et al. [16] in the CALPHAD calculation.

Experimental

Nd (99.99 % purity) and Mn (99.99 % purity) were used as the starting materials. Ingots with the mass of about 3 g were melted three times in an arc furnace by arc-melting pure elements using a nonconsumable tungsten electrode under an inert argon atmosphere. No chemical analysis of the alloys was conducted because the mass loss was generally less than 1 % during arc-melting. To achieve compositional homogeneity, the alloys were wrapped in tantalum foil and sealed in evacuated quartz tubes under

vacuum to be annealed at 773 K for 720 h, respectively, and then quenched in ice water.

Thermal analysis measurements of the annealed and as-cast alloys were investigated in Al_2O_3 crucibles under a flowing pure Ar atmosphere using differential scanning calorimeter (DSC, Netzsch STA 449 F3) and differential thermal analysis (DTA, TA Instruments SDT/Q-600) at different heating and cooling rates of 5 and 10 K min^{-1}, respectively. Thermal analysis measurements were calibrated using calibration materials In, Sn, Bi, Zn, Al, Ag, Au and Ni (TA Instruments-Waters LLC) as standard samples in order to eliminate the random errors and systematic errors. The overall uncertainty of the determined phase transformation temperature was estimated to be ±1 K.

Thermodynamic models

Pure elements

In the Mn–Nd binary system, the Gibbs energy for the element i in ϕ status is given as:

$$\begin{aligned} {}^0G_i^{\varphi}(T) &= G_i^{\varphi}(T) - H_i^{SER} \\ &= a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 \\ &\quad + hT^{-9} \end{aligned} \tag{1}$$

where H_i^{SER} is the enthalpy of the element i in its standard reference state (SER) at 298.15 K and 1 bar; T is the absolute temperature in K; $G_i^{\phi}(T)$ is the Gibbs energy of the element i with structure ϕ; ${}^0G_i^{\phi}(T)$ is the molar Gibbs energy of the element i with the structure of ϕ referred to the enthalpy of its stable state at 298.15 K and 1 bar. In the present work, ${}^0G_{Mn}^{\phi}(T)$ and ${}^0G_{Nd}^{\phi}(T)$ are taken from the Scientific Group Thermodata Europe (SGTE) database compiled by Dinsdale [26].

Solution phases

In the Mn–Nd system, the solution phase ϕ (ϕ = liquid, α-Mn(cbcc_A12), β-Mn(cub_A13), γ-Mn(fcc_A1), δ-Mn(bcc_A2) and α-Nd(dhcp), β-Nd(bcc_A2)) is described with the substitutional solution model based on random mixing of the constituent atoms. The Gibbs energies of these solutions are expressed as follows:

$$\begin{aligned} G_m^{\varphi} &= \sum_{i=Mn,Nd} x_i^0 G_i^{\varphi} + RT \sum_{i=Mn,Nd} x_i \ln x_i \\ &\quad + x_{Mn}x_{Nd} \sum_{j=0}^{n} {}^jL_{Mn,Nd}^{\varphi}(x_{Mn} - x_{Nd})^j \end{aligned} \tag{2}$$

where ${}^0G_i^{\phi}$ is the molar Gibbs free energy of pure component i in the respective reference state with the structure ϕ in a nonmagnetic state, which is taken from the compilation by Dinsdale [26]. x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; ${}^jL_{Mn,nd}^{\varphi}$ is the interaction parameter between elements Mn and Nd atoms and is expressed as:

Table 1 Experimental results of Mn–Nd alloys measured by thermal analysis

Sample composition/at.%	Phase transformation temperature/K				
	Invariant reactions				Liquidus
Mn10Nd90	967.4	981.5			1153.8
Mn15Nd85	968.7	981.4			1084.9
Mn20Nd80 (as-cast)	967.3	980.1			1015.4
Mn25Nd75	967.7				–
Mn30Nd70	966.1				995.4
Mn35Nd65	968.7				1047.5
Mn40Nd60	968.4				1082.1
Mn45Nd55	967.9				1117.1
Mn50Nd50	968.6		1113.5		1153.2
Mn55Nd45	968.2		1114.1		1174.6
Mn60Nd40 (as-cast)	969.1		1116.8		1185.8
Mn70Nd30	968.8[a]		1116.4	1210.7	1237.7
Mn75Nd25 (as-cast)	969.4[a]		1118.1	–	1282.2
Mn80Nd20	971.1[a]		–	1213.8	–

[a] The temperatures of the invariant reaction (L ↔ Mn_2Nd + α-Nd) were also measured by thermal analysis in Mn70Nd30, Mn75Nd25 and Mn80Nd20 alloys, resulting from nonequilibria microstructure of the alloys, even if the Mn70Nd30 and Mn80Nd20 alloys were annealed at 773 K for 720 h

$$^{j}L^{\varphi}_{\mathrm{Mn,Nd}} = A_{\mathrm{j}} + B_{\mathrm{j}}T \quad (3)$$

where A_{j} and B_{j} are parameters to be optimized in the present work.

Note that the lattice stabilities of Mn in α-Nd(dhcp) structure are not available in the SGTE database given by Dinsdale [26]. In the present work, the molar Gibbs free energy of Mn in α-Nd(dhcp) structure ($^{0}G^{\mathrm{dhcp}}_{\mathrm{Mn}}$) is taken from the data reported by Wang [27] as given:

$$^{0}G^{\mathrm{dhcp}}_{\mathrm{Mn}} = 10,000 + {}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} \quad (4)$$

Intermetallic compounds

According to the experimental information mentioned above, the intermetallic compounds, Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$, are stable in the Mn–Nd binary system and are treated as stoichiometric compounds in this work due to the lack of heat capacity data. Using the Neumann–Kopp rule, the molar Gibbs energies of the intermetallic compounds (Mn_aNd_b) are expressed as follows:

$$G^{\mathrm{Mn}_a\mathrm{Nd}_b}_{\mathrm{m}} = a^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + b^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + C_{\mathrm{i}} + D_{\mathrm{i}}T \quad (5)$$

where the parameters C_{i} and D_{i} are to be optimized in the present work.

Results and discussion

Experimental results

In the present work, fourteen Mn–Nd alloys were measured by means of thermal analysis (DTA and DSC). Different heating/cooling rates (5 and 10 K min^{-1}) were attempted to obtain the optimum rate in the DTA and DSC measurements. Generally, 10 K min^{-1} was used for most alloys in the present experiments, while 5 K min^{-1} was taken for the $Mn_{20}Nd_{80}$, $Mn_{25}Nd_{75}$ and $Mn_{50}Nd_{50}$ alloys because the compositions of these alloys are closer to the liquid composition of the invariant reactions. The onset temperature of the thermal effect in the heating curves was

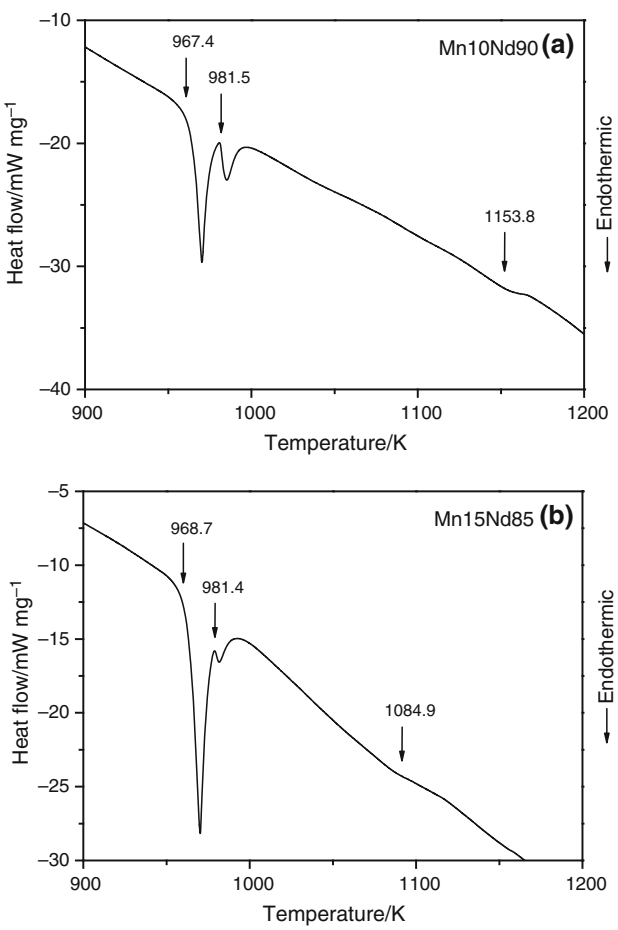

Fig. 1 Thermal analysis curves of the annealed Mn–Nd alloys at 10 K min^{-1} heating rate. **a** $Mn_{10}Nd_{90}$ alloy and **b** $Mn_{15}Nd_{85}$ alloy

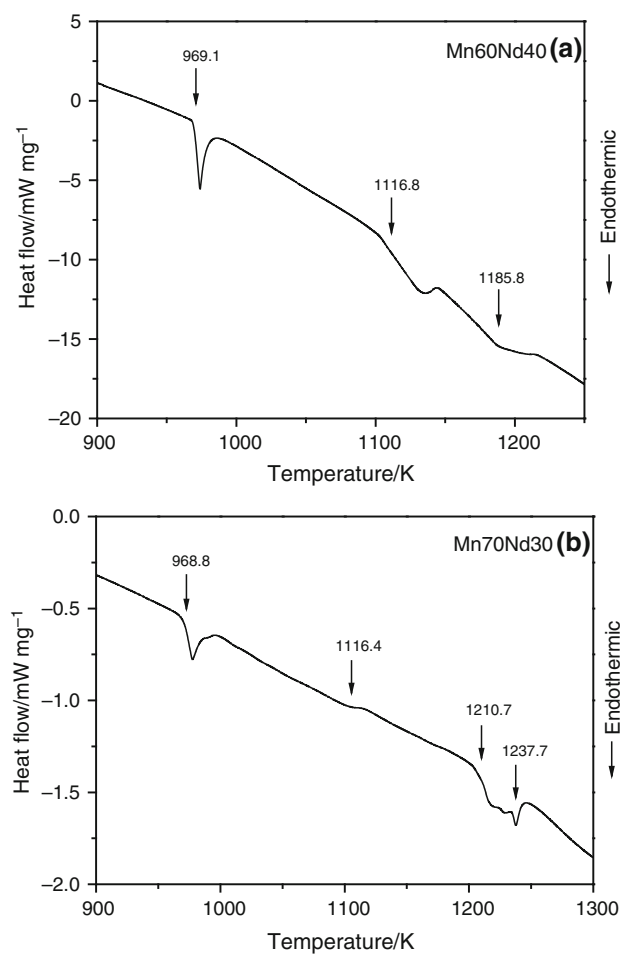

Fig. 2 Thermal analysis curves of the Mn–Nd alloys at 10 K min^{-1} heating rate. **a** $Mn_{60}Nd_{40}$ alloy (as-cast) and **b** $Mn_{70}Nd_{30}$ alloy (annealed)

selected as the temperature of the invariant reaction, while the peak temperature of the last thermal effect was used as the liquidus temperature. In the present work, phase transformation temperatures of the Mn–Nd alloys determined by DTA and DSC measurements are summarized in Table 1.

Figures 1 and 2 show the representative thermal analysis curves of the as-cast and annealed Mn–Nd alloys measured in the present work. As can be seen in Fig. 1, there are three peaks in the thermal analysis curves of the annealed $Mn_{10}Nd_{90}$ and $Mn_{15}Nd_{85}$ alloys, which are responsible to be the eutectic reaction (L ↔ Mn_2Nd + α-Nd) at 967.4 and 968.7 K, the catatectic reaction (α-Nd ↔ L + β-Nd) at 981.5 and 981.4 K and their liquidus temperatures at 1153.8 K and 1084.9 K, respectively. Figure 2 is the thermal analysis curves of the as-cast $Mn_{60}Nd_{40}$ alloy and annealed $Mn_{70}Nd_{30}$ alloy. The temperatures of the eutectic reaction (L ↔ Mn_2Nd + α-Nd) at 969.1 and 968.8 K and the peritectic reaction (L + $Mn_{23}Nd_6$ ↔ Mn_2Nd) at 1116.8 and 1116.4 K were measured, respectively, while the temperatures of the peritectic reaction (L + $Mn_{17}Nd_2$ ↔ $Mn_{23}Nd_6$) were determined to be 1210.7 K in Fig. 2b. Meanwhile, the liquidus temperatures of these two alloys are 1185.8 and 1237.7 K, respectively.

According to the DTA and DSC measurements of the Mn–Nd alloys, the experimental results are given in Table 1. In the present work, the temperatures of four invariant reactions in the Mn–Nd binary system were determined to be 968.4 K (L ↔ Mn_2Nd + α-Nd), 981.0 K (α-Nd ↔ L + β-Nd), 1115.8 K (L + $Mn_{23}Nd_6$ ↔ Mn_2Nd) and 1212.3 K (L + $Mn_{17}Nd_2$ ↔ $Mn_{23}Nd_6$), respectively. On the other hand, it should be mentioned that the temperatures of two eutectoid reactions (Mn_2Nd ↔ $Mn_{23}Nd_6$ + α-Nd and $Mn_{23}Nd_6$ ↔ $Mn_{17}Nd_2$ + α-Nd) could not be determined in the present work due to the difficulty in detecting the solid–solid phase transformation temperature [28].

Thermodynamic calculation

Using the stabilities of the elements Mn and Nd compiled by Dinsdale [26], the model parameters of various phases in the Mn–Nd binary system were optimized using the PARROT module in the Thermo-calc® software package developed by Sundman et al. [29] based on the experimental data determined in the present work and reported in the literature. This module works by minimizing the square sum of the differences between the experimental data and calculated values. In the optimization procedure, each set of the experimental data is given a certain mass according to the reliability and compatibility of the experimental data. It should be claimed that the masses of much more reliable experimental data are larger than those of less ones during the optimization. As for phase diagram data, it is better to use the experimental data measured by thermal analysis method. The mass can be changed systematically during the optimization until most of the experimental data are accounted for within the claimed uncertainty limits. Thermodynamic parameters of all condensed phases in the Mn–Nd binary system used and obtained finally in the present work are summarized in Table 2. The calculated temperatures and compositions of all invariant reactions in this binary system are given in Table 3.

Table 2 Thermodynamic parameters of the Mn–Nd binary system

Phase	Thermodynamic parameters	References
Liquid	${}^{(0)}L^{\mathrm{Liq}} = 7653.09 - 7.004T$	This work
	${}^{(1)}L^{\mathrm{Liq}} = -8641.69 + 5.619T$	
	${}^{(2)}L^{\mathrm{Liq}} = -7895.42 + 4.985T$	
δ-Mn,Nd (bcc)	${}^{(0)}L^{bcc}_{\mathrm{Mn,Nd}} = 25,267.94$	This work
	${}^{(1)}L^{bcc}_{\mathrm{Mn,Nd}} = 11,265.73$	
α-Nd(dhcp)	${}^{0}G^{\mathrm{dhcp}}_{\mathrm{Mn}} = {}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + 10,000$	[25]
	${}^{(0)}L^{\mathrm{dhcp}}_{\mathrm{Mn,Nd}} = 12,850$	This work
$Mn_{17}Nd_2$	$G^{\mathrm{Mn_{17}Nd_2}}_{\mathrm{Mn:Nd}} = -50,000 - 3.069T + 17{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + 2{}^{0}G^{\mathrm{hcp}}_{\mathrm{Nd}}$	This work
$Mn_{23}Nd_6$	$G^{\mathrm{Mn_{23}Nd_6}}_{\mathrm{Mn:Nd}} = -6678.31 - 85.572T + 23{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + 6{}^{0}G^{\mathrm{hcp}}_{\mathrm{Nd}}$	This work
Mn_2Nd	$G^{\mathrm{Mn_2Nd}}_{\mathrm{Mn:Nd}} = 7018.54 - 16.454T + 2{}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Nd}}$	This work
γ-Mn(fcc)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Mn}}$ cited from SGTE database	[24]
β-Mn(cub)	${}^{0}G^{\mathrm{cub}}_{\mathrm{Mn}}$ cited from SGTE database	[24]
α-Mn(cbcc)	${}^{0}G^{\mathrm{cbcc}}_{\mathrm{Mn}}$ cited from SGTE database	[24]

Table 3 Invariant reactions in the Mn–Nd binary system

Reactions	Type	T/K	Composition/x_{Mn}			References
L ↔ Mn_2Nd + α-Nd	Eutectic	973	–	0.667	–	[16]
		968	0.27	0.667	–	[19]
		958	0.278	0.667	0.023	[14]
		968.4	–	–	–	This work (Exp.)
		968.1	0.261	0.667	0.039	This work (Cal.)
L + $Mn_{23}Nd_6$ ↔ Mn_2Nd	Peritectic	1193	–	0.793	0.667	[16]
		1093	0.512	0.793	0.667	[19]
		1123	0.525	0.791	0.660	[14]
		1116.4	–	–	–	This work (Exp.)
		1116.0	0.509	0.793	0.667	This work (Cal.)
L + $Mn_{17}Nd_2$ ↔ $Mn_{23}Nd_6$	Peritectic	1213	0.676	0.895	0.791	[14]
		1213.8	–	–	–	This work (Exp.)
		1212.8	0.688	0.895	0.793	This work (Cal.)
L + β-Mn ↔ $Mn_{17}Nd_2$	Peritectic	1298	0.781	0.999	0.895	[14]
		1297.6	0.839	0.999	0.895	This work (Cal.)
δ-Mn ↔ L + γ-Mn	Catatectic	1407	–	–	–	[13]
		1411	–	–	–	[14]
		1384.6	0.908	0.997	0.999	This work (Cal.)
L + γ-Mn ↔ β-Mn	Peritectic	1360	–	–	–	[13]
		1373	–	–	–	[14]
		1360.0	0.890	0.999	0.999	This work (Cal.)
β-Nd ↔ L + α-Nd	Catatectic	1135	–	–	–	[16]
		1001	0.05	0.221	0.023	[14]
		981.4	–	–	–	This work (Exp.)
		981.0	0.077	0.245	0.036	This work (Cal.)
Mn_2Nd ↔ $Mn_{23}Nd_6$ + α-Nd	Eutectoid	~853	0.667	0.793	0.001	[19]
		~873	0.660	0.791	0.023	[14]
		853.0	0.667	0.793	0.027	This work (Cal.)
$Mn_{23}Nd_6$ ↔ $Mn_{17}Nd_2$ + α-Nd	Eutectoid	~753	0.793	0.895	0.001	[19]
		~773	0.791	0.895	0.023	[14]
		753.0	0.793	0.895	0.017	This work (Cal.)

Figure 3 is the calculated Mn–Nd phase diagram in the present work. The comparison of the calculated phase diagram with the experimental data determined in the present work and reported by Kirchmayr [18], Mostafa et al. [16] and Saccone et al. [21] is shown in Fig. 4. Generally, the calculated liquidus is in good agreement with the experimental data [16, 21] and the calculated results [15, 16]. It should be mentioned that the calculated liquidus in composition range of about 40–80 at.% Mn shows slightly lower than the experimental data determined in present work. The calculated liquidus at the Mn-rich side is obviously lower than the experimental data determined by Kirchmayr and Lugscheider [18], while it is in good agreement with the experimental data determined in present work and in Refs. [16, 21]. Therefore, the calculated liquidus in the present work is reasonable and acceptable. In combination with Table 3, the calculated temperatures and compositions of the invariant reactions associated with the liquid phase agree well with the experimental data determined in the present work and the calculated results in Refs. [16, 21] within experimental error. However, It should be pointed out that the temperature of the invariant reactions, L ↔ Mn_2Nd + α-Nd, Mn_2Nd ↔ $Mn_{23}Nd_6$ + α-Nd and $Mn_{23}Nd_6$ ↔ $Mn_{17}Nd_2$ + α-Nd, was calculated to be 968.1, 853.0 and 753.0 K, which are closer to the experimental results determined by Saccone et al. [21]. The calculated temperatures of the catatectic reaction (β-Nd ↔ L + α-Nd, at 981.0 K) and the peritectic reaction (L + $Mn_{23}Nd_6$ ↔ Mn_2Nd, at 1116.0 K) agree well with the experimental

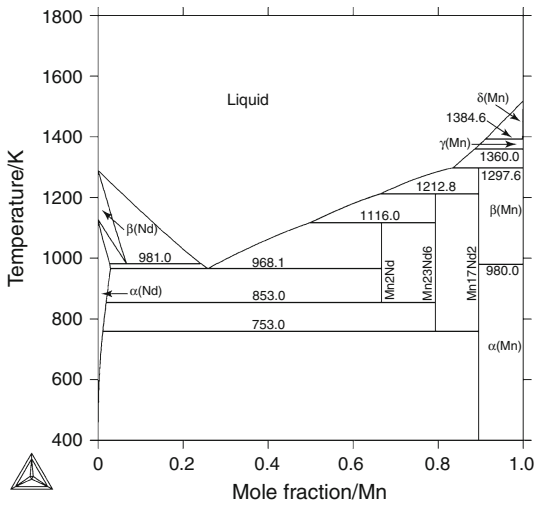

Fig. 3 Calculated phase diagram of the Mn–Nd binary system in the present work

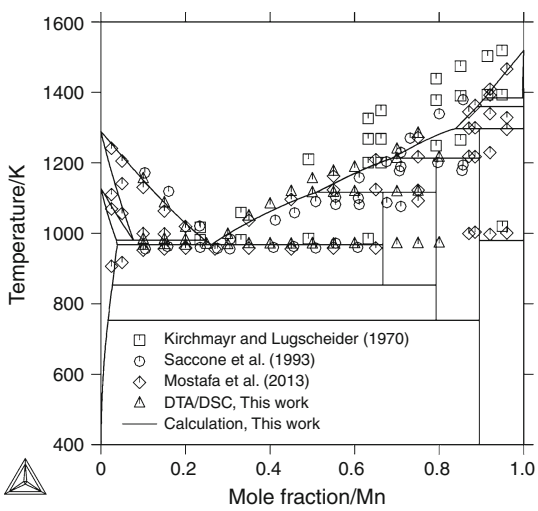

Fig. 4 Calculated phase diagram of the Mn–Nd system with the experimental data determined in the present work and reported in Refs. [14, 16, 19]

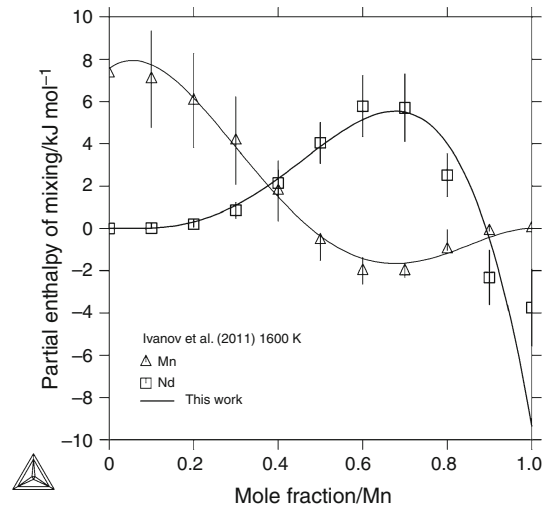

Fig. 5 Calculated partial enthalpy of Mn and Nd in liquid alloys at 1600 K with experimental data [23] (reference states: liquid Mn and liquid Nd)

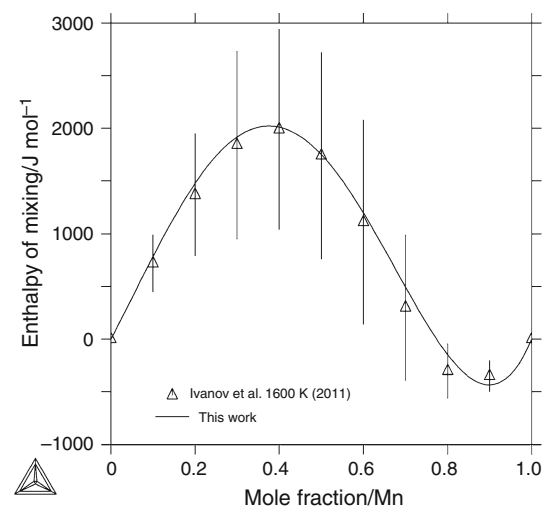

Fig. 6 Calculated enthalpy of mixing of the liquid Mn–Nd alloys at 1600 K with the experimental data [23] (reference states: liquid Mn and liquid Nd)

results measured in this work, which are much lower than the experimental data determined by Mostafa et al. [16]. In addition, as can be seen in Fig. 4 and Table 3, the calculated solubility of Mn in α-Nd is 3.9 at.% Mn at 968.1 K, while that of Mn in β-Nd is 7.7 at.% Mn at 981.0 K. It is indicated that the present calculated results agree well with the experimental data [16]. In addition, it should be mentioned that the calculated temperature of the catatectic reaction (δ-Mn ↔ L + γ-Mn at 1384.6 K) is lower (22 and 26 K) than the reported values in Refs. [15] and [16]. The reason for it could be that the solid solubilities of Nd in δ-Mn and γ-Mn were not taken into consideration due to the lack of experimental data. Thermodynamic parameters of δ-Mn and γ-Mn phases could not be fixed, resulting in the difference of this catatectic reaction temperature between the calculated value and the reported values in Refs. [15, 16].

Figure 5 shows the calculated partial molar enthalpies of Mn and Nd in liquid alloys at 1600 K with the experimental data [25]. It can be seen that the calculated results are satisfactorily consistent with the experimental data within the estimated error. However, the calculated partial enthalpies of Mn and Nd in liquid alloys at 1600 K by Kim and Jung [15] and Mostafa et al. [16] show much more positive deviation in the composition range of 50–90 at.% Mn. The comparison of the enthalpy of mixing of the liquid Mn–Nd alloys at 1600 K between the calculated results

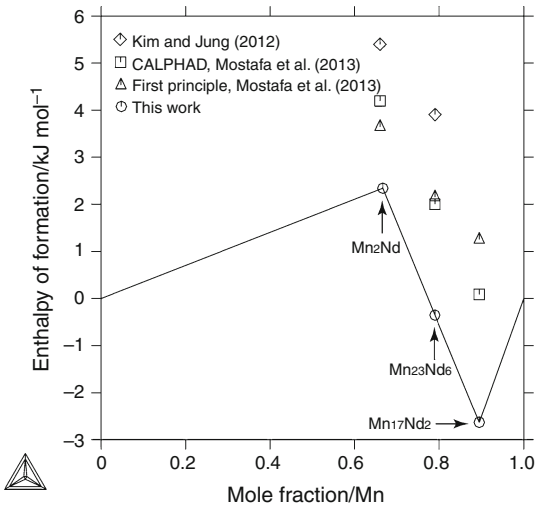

Fig. 7 Calculated enthalpies of formation of the intermetallic compounds (Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$) at 298 K with the reported values [13, 14] (reference states: α-Mn and α-Nd)

and experiment values is given in Fig. 6. The calculated mixing enthalpy is in good agreement with the experimental data measured by Ivanov et al. [25]. It should be pointed out that the enthalpy of mixing of the liquid Mn–Nd alloys calculated by Mostafa et al. [16] is in good agreement with the experimental data [25] at Nd-rich side, but shows slight deviation at Mn-rich side.

Figure 7 is the calculated enthalpies of formation of the intermetallic compounds in the Mn–Nd system at 298 K. The enthalpies of formation of the intermetallic compounds, Mn_2Nd, $Mn_{23}Nd_6$ and $Mn_{17}Nd_2$, were calculated to be 2339.51, −351.49 and −2631.58 J mol^{-1} in the present work, respectively. Due to the decomposition at low temperature (seen in Fig. 3), the enthalpies of formation of $Mn_{23}Nd_6$ and Mn_2Nd would be positive or less negative, while that of stable $Mn_{17}Nd_2$ at room temperature is negative. However, the values calculated by Kim and Jung [15] and Mostafa et al. [16] are all positive through the first-principle calculations and CALPHAD method. In addition, Boer et al. [30] indicated that the enthalpies of formation of the intermetallic compounds in the Mn–RE binary systems are between −1000 and −2000 J mol^{-1}. Hence, the calculated results in the present work are reasonable. It is very necessary that the enthalpies of formation of the intermetallic compounds in the Mn–Nd binary system should be measured in the further experimental investigations.

Conclusions

The phase diagram of the Mn–Nd binary system was measured through fourteen alloys prepared by arc-melting method with differential scanning calorimeter and differential thermal analysis. The temperatures of the liquidus and invariant reactions in the Mn–Nd binary system were determined in the present work. On the basis of experimental thermochemical data and phase diagram data in the present work and published literature, the Mn–Nd binary system was thermodynamically re-optimized using the CALPHAD method. A set of self-consistent thermodynamic parameters has been obtained by critically evaluated literature data. The calculated phase diagram and thermodynamic properties agree well with the experimental data.

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